

Walk-unbiased doubly robust graph propagation for recommendation from exposure-biased logs

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to support code development and to polish the writing. All ideas and results are the author's own. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

All data sets are public: Coat (<https://www.cs.cornell.edu/~schnabts/mnar/>), Yahoo! R3 (Yahoo! Webscope; the version distributed with the AutoDebias code, <https://github.com/DongHande/AutoDebias>) and KuaiRec 2.0 (<https://zenodo.org/records/18164998>).

Code, configurations and all result files are available at https://github.com/vinhqdang/causal_learning_graph_recommendation.